

# Demonstrating AHA: Boosting Unmodified AI's Robustness by Proactively Inducing Favorable Human Sensing Conditions

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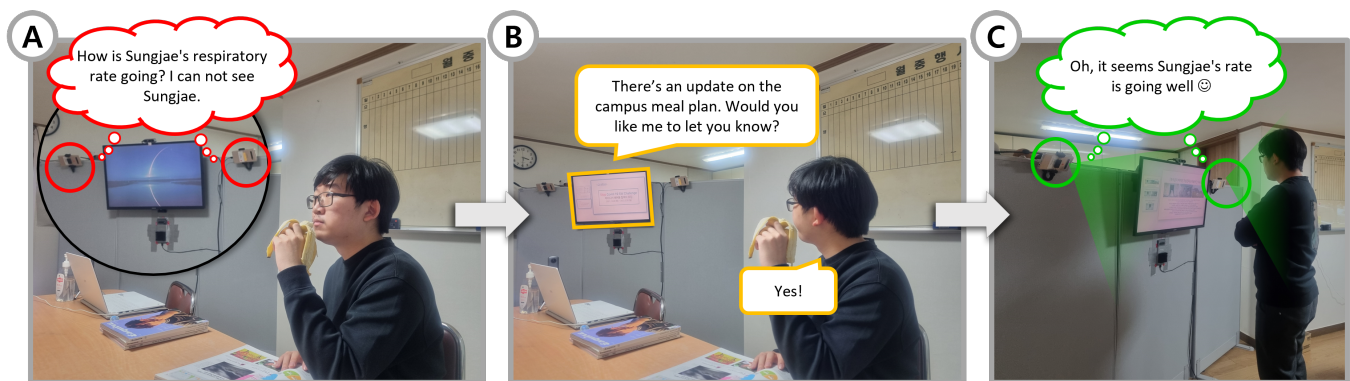
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**Figure 1:** An example of how AI-to-Human Actuation complements an AI sensor under perception difficulty: (A) two vision sensors suffering an unfavorable condition, i.e., unable to locate the target person from their FoVs. (B) Content push on a proxemic display [7] triggered as an actuation to re-locate the target person. (C) The unfavorable condition has been resolved by the human reaction and interaction, i.e., relocation of the target human

## ABSTRACT

Imagine a near-future smart home. Home-embedded visual AI sensors continuously monitor the resident, inferring her activities and internal states that enable higher-level services. Here, as home-embedded sensors passively monitor a free person, good inferences happen inconsistently. The inferences' confidence highly depends on how congruent her momentary conditions are to the conditions favored by the AI models, e.g., front-facing or unobstructed.

We envision new strategies of AI-to-Human Actuation (AHA) that boost the sensory AI's robustness by inducing favorable conditions from the person with proactive actuations. To demonstrate our concept, in this demo, we show how the inference quality of the AI model changes relative to the person's conditions and introduce

possible actuations, used in our full paper experiments, that could drive more favorable conditions for visual AIs.

## CCS CONCEPTS

• **Human-centered computing** → Empirical studies in interaction design; Ambient intelligence; • **Computing methodologies** → Intelligent agents; Active vision.

## KEYWORDS

Human-AI Interaction; IoT; Actuation

### ACM Reference Format:

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## 1 INTRODUCTION

Imagine your toddler son is playing on the floor. You just see that he is taking something into his mouth but you cannot confirm what that is as he is sitting sideways to you. You hail him; he turns

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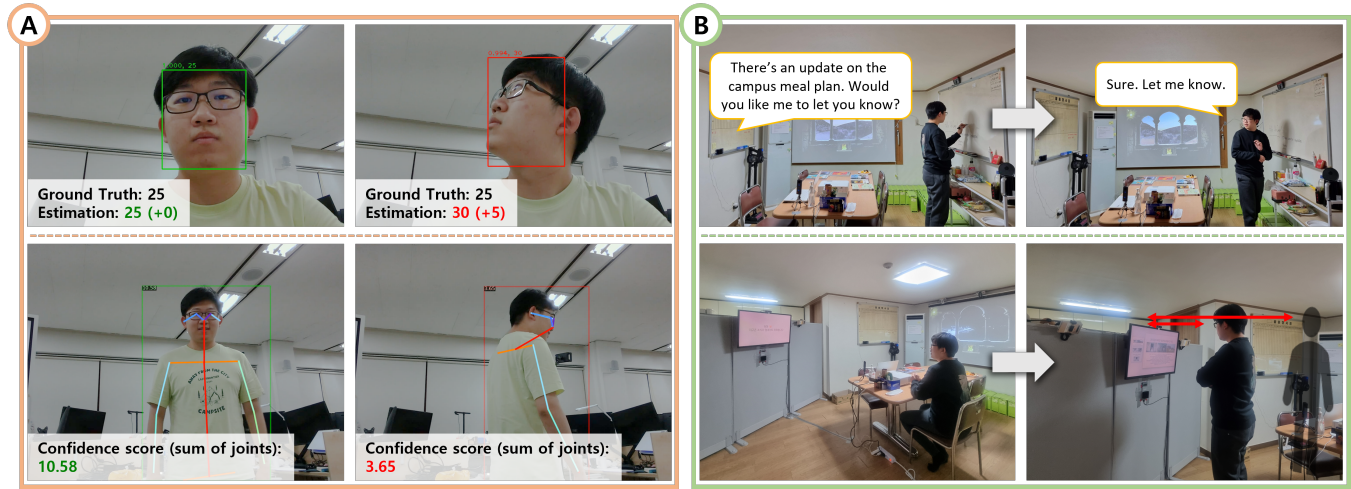
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**Figure 2: Demonstration plan: (A) error varying by the facial orientation in the age estimation model [2] and confidence score varying by the body orientation in the pose estimation model [32] (B) actuation embodiments to show in the demo – a smart speaker and an interactive display that was used in the full paper [5]**

around. Now you see a clear view of a leftover cookie in his hand, feeling relaxed. Similarly, a kindergarten kid has just rushed to her grandfather, showing off her masterpiece drawing right in front of his eyes, as usual. The grandfather, farsighted, cannot see the details. “Wow, too close, sweetie.” he says. “Oops, sorry grandpa.” she moves it a little back.

Like the above situations, our human-to-human perception abilities are often challenged by unfavorable conditions, and we often overcome such challenges by actively actuating the target towards a more favorable condition to help our perception, rather than passively waiting until the conditions become favorable.

In this light, we proposed a novel strategy *AI-to-Human Actuation* (AHA) [5] which applies proactive actuations on the target human to induce her conditions more favorable to AIs. Through AHA, AI-powered sensors that continuously monitor the target human [10, 11, 19, 20, 23, 26, 35] can be complemented and become robust towards unfavorable conditions that happen during its inference.

Figure 1 depicts an example scenario of AHA out of many that we developed. Two vision AI sensors continuously infer the target human’s health conditions, e.g., respiratory rate, by observing his upper body. However, the target human is out of the FoV of two sensors, making their inference difficult. At this point, a proactive actuation is given, e.g., a content push from the near proxemic display. It naturally induces him to move towards the display, yielding a view favorable to the vision AI sensor.

In this demo, we demonstrate the motivation of AHA and the actuation embodiments for AHA. Specifically, we show that visual AI models have favorable and unfavorable conditions to sense the target human, and introduce a few embodiments activated in smart devices, which were also used in our full paper [5], that could handle such unfavorable conditions.

## 2 RELATED WORKS

Broadly, AHA is positioned in the problem space of facilitating human-targeted AI to be more effective in unfavorable real-life sensing conditions. Approaches for robust AI in ML, CV, and HCI literature [4, 8, 36] mostly increased the complexity of the AI model or system, making it harder to run on resource-tight, IoT platforms.

Unlike previous approaches, AHA adopts proactive interactions to boost the AI’s robustness. Proactive interactions have been extensively studied in the HCI community, including push notifications [12, 14, 16, 24, 25, 27, 28], proactive speakers [3, 17, 31], home robotics & automation [13, 15, 18], gamifications [6, 9, 21], and interactive displays [1, 30, 33, 34]. These interactions engage humans to react and change the user’s state (e.g., start speaking, move towards the device). Consequently, the trend of proliferating proactive interactions shows the readiness of AHA, extending the pool of interactions that could be used as actuations.

The spectrum of human-AI interaction has been largely discussed in terms of the balance between human control and computer automation [22, 29]. Mostly, these models converged in assisting human tasks. Although AHA shares the key stakeholders (i.e., humans, AIs, and signals in between), AHA *assists the AIs* to perform better by deploying actuation but remains *as transparent as possible* to the human.

## 3 AI-TO-HUMAN ACTUATION

Abstractly, three steps are needed to embody and evaluate AHA in a living space.

### 3.1 Perceiving AI’s Unfavorable Conditions

Unfavorable conditions that the AI model currently suffers should be first investigated. Recognizing such information hints at selecting the type and variables of actuations that can help the AI model overcome those conditions. For example, whether the target human face is viewed sideways, and if so at what direction and angle

the face is turned (e.g., left, 30°), or whether the face is hidden are possible information that could be determined in the case of a face detection model. This information helps in selecting effective actuations for each unfavorable condition, e.g., actuations triggered from the human's right, left, or back.

### 3.2 Matching between Unfavorable Conditions and Existing Actuations

In a living space, there are numerous installed sensors running an AI model and numerous actuators that are capable of triggering proactive actuations. Here, there should be a method or an algorithm matching these two in an objective to relieve real-time happening AI's unfavorable conditions with the information from the first step. In our full paper experiment, the matching was done in a Wizard-of-Oz manner with the self-defined criteria to discover such methods. Each criterion was composed of (1) an unfavorable condition, (2) sensors that suffer from that condition, (3) a candidate actuation(s) to trigger, and (4) the expected human reaction.

### 3.3 Evaluation: Performance Improvement and User Acceptance

After the actuation is selected and triggered for the AI model, two perspectives should be considered in evaluating the strategy – performance improvement and user acceptance. The first perspective is to find out whether the actuation truly supported the AI model to overcome its unfavorable conditions. From this perspective, the improvement in AI inference quality and the time for the quality to surpass a moderate threshold can be considered. The second is to guarantee that the actuated person's experience is not negatively affected due to the proactivity of actuations.

In our full paper, we conducted an experiment with 20 participants to evaluate AHA. Each participant stayed at our constructed testbed for 1 hour and five types of actuations were given in a Wizard-of-Oz manner. After the 1 hour session, 30min 1-on-1 semi-structured interview followed to examine the user experience of each actuation type as well as the overall experience of the actuation-enabled environment. The experiment was approved by our university IRB.

From the experiment, we found that 92.1% actuations improved the AI model performance and such improvements happen mostly within 20-30 seconds. Furthermore, the trend of user experience to actuations was positive, perceiving actuation as a near-future smart home action and as a kind of care. For more detailed explanations and results, please refer to our full paper [5].

## 4 DEMONSTRATION PLAN

Our demo will be done in two-fold as shown in Figure 2. We first motivate people that unfavorable conditions can happen in real-time inference, and that favorable conditions exist in the AI model. Then, we introduce a number of actuations, so that people can try them, and experience how the actuation can derive AI's favorable conditions and how it feels to them when triggered.

Specifically, we plan to prepare a display to visualize the real-time inference quality of the visual AI models deployed at the demo venue (e.g., age estimation, pose estimation) where the live video

feed from the installed camera is given. With this setup, we guide people to find favorable and unfavorable conditions of the AI model, for example, by changing facial angle towards the camera, varying distance to the camera, or hiding behind obstacles.

We also plan to prepare a set of actuators (e.g., smart speakers, interactive displays) that could alleviate the AI models' unfavorable conditions that attendees have discovered earlier. We will introduce each actuator and trigger an actuation from the actuator, inducing people to respond and freely interact with it. The visitors to our demo should be able to have comprehensive experiences that help them assess the degree of noticeability of actuation unawaresly applied to them, the degree of AI performance improvement, and the degree of positive or negative influence along with their natural routine.

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